P1 $\boxed{4.5}$

$$Z = 1 + e^{-\beta\Delta}\left(e^{\beta g\mu_B B} + 1 + e^{-\beta g\mu_B B}\right)$$
$$= e^{-\beta\Delta}\left[e^{\beta\Delta} + e^{\beta g\mu_B B} + 1 + e^{-\beta g\mu_B B}\right]$$
$$F = -nkT \ln Z \qquad = e^{-\beta\Delta}\left[1 + e^{\beta\Delta} + 2\cosh \beta g\mu_B B\right]$$
$$= -nkT\left[-\beta\Delta + \ln\left[1 + e^{\beta\Delta} + 2\cosh \beta g\mu_B B\right]\right]$$

$$M = -\left(\frac{\partial F}{\partial B}\right)_T = n\cancel{kT}\,\frac{2\sinh \beta g\mu_B B \cdot \cancel{\beta} g\mu_B}{1 + e^{\beta\Delta} + 2\cosh \beta g\mu_B B}$$

$$M = \frac{2n\beta(g\mu_B)^2 B}{3 + e^{\beta\Delta}}$$

If $\chi \ll 1$, $\quad B \approx \mu_0 H$

$$\therefore \quad \chi \equiv \frac{M}{H} \simeq \frac{\mu_0 M}{B} = \frac{2n\mu_0 g^2\mu_B^2}{k_B T\left(3 + e^{\Delta/k_B T}\right)}$$

Note that at high temperature $\quad k_B T \gg \Delta$

$e^{\Delta/k_B T} \to 1$ so that $\chi \to \dfrac{n\mu_0 g^2\mu_B^2}{2k_B T}$,

which is a Curie law, whether $\Delta > 0$ or $\Delta < 0$.

At low temperature $\quad k_B T \ll \Delta$, then if

$\Delta > 0 \quad \chi \to 2n\mu_0 g^2\mu_B^2 \beta\, e^{-\Delta/k_B T}$

$\to 0$ but if $\Delta < 0 \quad \chi \to 2n\mu_0 g^2\mu_B^2/3k_B T.$

$\because$ non-magnetic $\Rightarrow \chi \to 0$ only singlet occupied.

It corresponds to all levels occupied and $\frac{3}{4}(1)(1+1) + \frac{1}{4}\cdot 0 = \frac{3}{2} \times$ a classical Curie law.

It corresponds to just the triplet occupied and so $1(1+1) = 2 \times$ a classical Curie law.

(P2)

$\boxed{4.7}$ $\mathcal{H} = \mathcal{H}_1 + \mathcal{H}_2 + \mathcal{H}'$

operate on $\Psi_\pm$.

$$\langle \psi_\pm | \mathcal{H}_1 | \psi_\pm \rangle = E_1$$
$$\langle \psi_\pm | \mathcal{H}_2 | \psi_\pm \rangle = E_2$$

$$\langle \psi_\pm | \mathcal{H}' | \psi_\pm \rangle$$

$$= \frac{1}{2}\Big(\big[\langle \psi_a(r_1)\psi_b(r_2)| \pm \langle \psi_a(r_2)\psi_b(r_1)| \big] \mathcal{H}' \big[|\psi_a(r_1)\psi_b(r_2)\rangle \pm |\psi_a(r_2)\psi_b(r_1)\rangle \big]$$

(divide by $\frac{1}{2}\big[\langle \psi_a(r_1)\psi_b(r_2) \pm \langle \quad | \big]\big[| \quad \rangle \pm | \quad \rangle \big]$

$= \frac{1}{2}\big[2 \pm 2S \big] = 1 \pm S$)

$$= \frac{\frac{1}{2}(2K \pm 2J)}{1 \pm S} = \frac{K \pm J}{1 \pm S} = \frac{\langle \psi_\pm | H' | \psi_\pm \rangle}{\langle \psi_\pm | \psi_\pm \rangle}$$

(Heitler-London approximation.)

P3 $\boxed{5.5}$ $\frac{M}{M_s} = \tanh y$ ①

$$y = \frac{\mu(\lambda M)}{kT} \Rightarrow \frac{M}{M_s} = \underbrace{\frac{k_B}{\mu \lambda M_s}}_{1/T_c} T y \quad ②$$

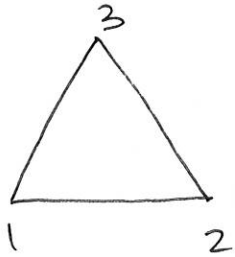
When $y \gg 1$, $\tanh y = \frac{e^y - e^{-y}}{e^y + e^{-y}} = \frac{1 - e^{-2y}}{1 + e^{2y}} \approx 1 - 2e^{-2y}$ ③

① & ② & ③ $\Rightarrow \frac{Ty}{T_c} \approx 1 - 2e^{-2y}$ $\therefore \frac{M}{M_s} \approx 1 - 2e^{-2\frac{T_c}{T}\left(\frac{M}{M_s}\right)}$

$\approx 1 - \ldots$

$$\approx \underline{1 - 2e^{-2T_c/T}}$$

P4. $H = J\left(\vec{S}_1\cdot\vec{S}_2 + \vec{S}_2\cdot\vec{S}_3 + \vec{S}_3\cdot\vec{S}_1\right) = \frac{1}{2}J\left[\left(\vec{S}_1+\vec{S}_2+\vec{S}_3\right)^2 - 3\vec{S}_1^{\,2}\right]$



$$= \frac{1}{2}J\left(\vec{S}_{tot}^{\,2} - 3\cdot\frac{3}{4}\right)$$

$$\vec{S}_{tot} = \vec{S}_1 + \vec{S}_2 + \vec{S}_3$$

Since $s = \frac{1}{2}$, $S_{tot} = \begin{cases} 3/2 \\ 1/2 \end{cases}$

$$\Rightarrow S_{tot} = \frac{3}{2}, \quad E_{3/2} = \frac{1}{2}J\left(\frac{3}{2}\left(\frac{3}{2}+1\right) - \frac{9}{4}\right) = \frac{3}{4}J$$

$$S_{tot} = \frac{1}{2}, \quad E_{1/2} = \frac{1}{2}\left(\frac{1}{2}\left(1+\frac{1}{2}\right) - \frac{9}{4}\right) = -\frac{3}{4}J$$

Eigenstates: start with fully polarized state

$S_{tot} = \frac{3}{2}$: $\left|\frac{3}{2},\frac{3}{2}\right\rangle = |\uparrow\uparrow\uparrow\rangle$

$$\hat{S}^-_{tot}\left|\frac{3}{2},\frac{3}{2}\right\rangle = \sqrt{\frac{3}{2}\left(\frac{3}{2}+1\right) - \frac{3}{2}\left(\frac{3}{2}-1\right)}\left|\frac{3}{2},\frac{1}{2}\right\rangle = \sqrt{3}\left|\frac{3}{2},\frac{1}{2}\right\rangle$$

On the other hand $\hat{S}^- = \hat{S}^-_1 + \hat{S}^-_2 + \hat{S}^-_3$

$$\Rightarrow \left|\frac{3}{2},\frac{1}{2}\right\rangle = \frac{1}{\sqrt{3}}\left(|\downarrow\uparrow\uparrow\rangle + |\uparrow\downarrow\uparrow\rangle + |\uparrow\uparrow\downarrow\rangle\right)$$

$$\hat{S}^-_{tot}\left|\frac{3}{2},\frac{1}{2}\right\rangle = \sqrt{\frac{3}{2}\left(\frac{3}{2}+1\right) - \frac{1}{2}\left(\frac{1}{2}-1\right)}\left|\frac{3}{2},-\frac{1}{2}\right\rangle = 2\left|\frac{3}{2},-\frac{1}{2}\right\rangle$$

$$\Rightarrow 2\left|\frac{3}{2},-\frac{1}{2}\right\rangle = \left(S^-_1 + S^-_2 + S^-_3\right)\frac{1}{\sqrt{3}}\left(\downarrow\uparrow\uparrow + \uparrow\downarrow\uparrow + \uparrow\uparrow\downarrow\right)$$

$$= \frac{1}{\sqrt{3}}\Big(|\downarrow\downarrow\uparrow\rangle + |\downarrow\uparrow\downarrow\rangle + |\downarrow\downarrow\uparrow\rangle + |\uparrow\downarrow\downarrow\rangle$$

$$+ |\downarrow\uparrow\downarrow\rangle + |\uparrow\downarrow\downarrow\rangle\Big)$$ (each state appears twice)

$$\Rightarrow \left|\frac{3}{2},-\frac{1}{2}\right\rangle = \frac{1}{\sqrt{3}}\left(|\downarrow\downarrow\uparrow\rangle + |\uparrow\downarrow\downarrow\rangle + |\downarrow\uparrow\downarrow\rangle\right)$$

$$\hat{S}^-_{tot}\left|\tfrac{3}{2},-\tfrac{1}{2}\right\rangle = \sqrt{\tfrac{3}{2}\left(\tfrac{3}{2}+1\right) - \left(-\tfrac{1}{2}\right)\left(-\tfrac{1}{2}-1\right)}\ \left|\tfrac{3}{2},-\tfrac{3}{2}\right\rangle$$

$$= \sqrt{3}\ \left|\tfrac{3}{2},-\tfrac{3}{2}\right\rangle$$

But $\sqrt{3}\left|\tfrac{3}{2},-\tfrac{3}{2}\right\rangle = \frac{1}{\sqrt{3}}\left(|\downarrow\downarrow\downarrow\rangle + |\downarrow\downarrow\downarrow\rangle + |\downarrow\downarrow\downarrow\rangle\right)$

$$= \sqrt{3}\ |\downarrow\downarrow\downarrow\rangle$$

$\Rightarrow$ of course $\left|\tfrac{3}{2},-\tfrac{3}{2}\right\rangle = |\downarrow\downarrow\downarrow\rangle$

Thus we found 4 states from the $S_{tot} = \frac{3}{2}$ manifold.

There are 4 more states from $S_{tot} = \frac{1}{2} \Rightarrow$ 2 doublets:

Try $\left|\tfrac{1}{2},\tfrac{1}{2}\right\rangle = \frac{1}{\sqrt{3}}\left(|\downarrow\uparrow\uparrow\rangle + e^{i\varphi}|\uparrow\downarrow\uparrow\rangle + e^{i2\varphi}|\uparrow\uparrow\downarrow\rangle\right)$

Normalized. Must be $\left\langle\tfrac{3}{2},\tfrac{1}{2}\middle|\tfrac{1}{2},\tfrac{1}{2}\right\rangle = 0$

$$\Rightarrow \frac{1}{3}\left(1 + e^{i\varphi} + e^{i2\varphi}\right) = 0$$

$$e^{i\varphi}\left(e^{-i\varphi} + 1 + e^{i\varphi}\right) = 0 \Rightarrow \cos\varphi = -\frac{1}{2}$$

$$\Rightarrow \begin{cases} (a)\ \varphi = \frac{2\pi}{3} \\ (b)\ \varphi = \frac{4\pi}{3}\ \left(e^{i\frac{4\pi}{3}} = e^{i\frac{6\pi-2\pi}{3}} = e^{-i\frac{2\pi}{3}}\right) \\ \quad \text{same as } \varphi = -\frac{2\pi}{3} \end{cases}$$

$\Rightarrow$ got two states

$$\left|\tfrac{1}{2},\tfrac{1}{2},+\right\rangle = \frac{1}{\sqrt{3}}\left(|\downarrow\uparrow\uparrow\rangle + e^{i\frac{2\pi}{3}}|\uparrow\downarrow\uparrow\rangle + e^{i\frac{4\pi}{3}}|\uparrow\uparrow\downarrow\rangle\right)$$

$$\left|\tfrac{1}{2},\tfrac{1}{2},-\right\rangle = \frac{1}{\sqrt{3}}\left(|\downarrow\uparrow\uparrow\rangle + e^{-i\frac{2\pi}{3}}|\uparrow\downarrow\uparrow\rangle + e^{-i\frac{4\pi}{3}}|\uparrow\uparrow\downarrow\rangle\right)$$

Check that they are orthogonal:

$$\langle \tfrac{1}{2}, \tfrac{1}{2}, + | \tfrac{1}{2}, \tfrac{1}{2}, - \rangle = \frac{1}{3}\left(1 + e^{-i\frac{4\pi}{3}} + e^{-i\frac{8\pi}{3}}\right) =$$

$$= \frac{1}{3}\left(1 + e^{-i\frac{6\pi}{3}} \cdot e^{i\frac{2\pi}{3}} + e^{-i2\pi} \cdot e^{-i\frac{2\pi}{3}}\right) =$$

$$= \frac{1}{3}\left(1 + 2\cos\frac{2\pi}{3}\right) = 0 \quad \underline{\text{OK}}$$

States with $S^z = -\frac{1}{2}$ are obtained in usual way:

$$\hat{S}^-_{tot} \left| \tfrac{1}{2}, \tfrac{1}{2}; \pm \right\rangle = \sqrt{\tfrac{1}{2}\left(\tfrac{1}{2}+1\right) - \tfrac{1}{2}\left(\tfrac{1}{2}-1\right)} \left| \tfrac{1}{2}, -\tfrac{1}{2}; \pm \right\rangle$$

$$\Rightarrow \left| \tfrac{1}{2}, -\tfrac{1}{2}; \pm \right\rangle = \left(S_1^- + S_2^- + S_3^-\right) \frac{1}{\sqrt{3}} \Big(|\downarrow\uparrow\uparrow\rangle + e^{\pm i\frac{2\pi}{3}} |\uparrow\downarrow\uparrow\rangle$$

$$+ e^{\pm i\frac{4\pi}{3}} |\uparrow\uparrow\downarrow\rangle \Big) = \frac{1}{\sqrt{3}} \Big(\underline{|\downarrow\downarrow\uparrow\rangle} + |\downarrow\uparrow\downarrow\rangle$$

$$+ e^{\pm i\frac{2\pi}{3}} \left(\underline{|\downarrow\downarrow\uparrow\rangle} + \underline{\underline{|\uparrow\downarrow\downarrow\rangle}} \right) + e^{\pm i\frac{4\pi}{3}} \left(|\downarrow\uparrow\downarrow\rangle + \underline{\underline{|\uparrow\downarrow\downarrow\rangle}} \right)$$

$$= \frac{1}{\sqrt{3}} \Big(\left(1 + e^{\pm i\frac{2\pi}{3}}\right) |\downarrow\downarrow\uparrow\rangle + \left(e^{\pm i\frac{2\pi}{3}} + e^{\pm i\frac{4\pi}{3}}\right) |\uparrow\downarrow\downarrow\rangle$$

$$+ \left(1 + e^{\pm i\frac{4\pi}{3}}\right) |\downarrow\uparrow\downarrow\rangle \Big)$$

But $1 + e^{\pm i\frac{2\pi}{3}} = \underbrace{\left(1 + e^{\pm i\frac{2\pi}{3}} + e^{\pm i\frac{4\pi}{3}}\right)}_{= 1 + 2\cos\frac{2\pi}{3} = 0} - e^{\pm i\frac{4\pi}{3}}$

$$\Rightarrow \left| \tfrac{1}{2}, -\tfrac{1}{2}; \pm \right\rangle = \frac{1}{\sqrt{3}} \Big(-|\uparrow\downarrow\downarrow\rangle - e^{\pm i\frac{2\pi}{3}} |\downarrow\uparrow\downarrow\rangle$$

$$- e^{\pm i\frac{4\pi}{3}} |\downarrow\downarrow\uparrow\rangle \Big) = -\frac{1}{\sqrt{3}} \Big(|\uparrow\downarrow\downarrow\rangle + e^{\pm i\frac{2\pi}{3}} |\downarrow\uparrow\downarrow\rangle + e^{\pm i\frac{4\pi}{3}} |\downarrow\downarrow\uparrow\rangle \Big)$$

The physical meaning of $e^{\pm i \frac{2\pi}{3}}$ factors?

(A) Momentum! (both linear and angular, $\hat{L}^z$)

Indeed, $$\left|\tfrac{1}{2}, \tfrac{1}{2}; \lambda\right\rangle = \frac{1}{\sqrt{3}}\Big(|\underline{\downarrow}\uparrow\uparrow\rangle + e^{i\varphi_\lambda}|\uparrow\underline{\downarrow}\uparrow\rangle + e^{i2\varphi_\lambda}|\uparrow\uparrow\underline{\downarrow}\rangle\Big) = \frac{1}{\sqrt{3}}\sum_{n=1,2,3} e^{i\varphi_\lambda(n-1)}|n\rangle$$

where $|n\rangle$ denotes state with <u>down</u> spin in the $\underline{\underline{n}}$-th position.

$$\varphi_\lambda = \begin{cases} \varphi_+ = \frac{2\pi}{3} & \text{for } \lambda = + \\ \varphi_- = -\frac{2\pi}{3} & \text{for } \lambda = - \end{cases}$$ describes clockwise and counterclockwise motion along the triangle.

(B) It is also an eigenstate of <u>scalar chirality</u> operator $\hat{\chi}_{123} = \vec{S}_1 \cdot \vec{S}_2 \times \vec{S}_3$, see Wen, Wilczek, Zee Phys Rev B <u>39</u>, 11413 (1989)

Mixed product of three spins implies, classically, non-coplanar spin configuration such that the "volume", measured by $\hat{\chi}$, is finite.

Scalar chirality $\chi = \vec{S}_1 \cdot \vec{S}_2 \times \vec{S}_3$

$$\hat{\chi} = \begin{vmatrix} S_1^x & S_1^y & S_1^z \\ S_2^x & S_2^y & S_3^z \\ S_3^x & S_3^y & S_3^z \end{vmatrix} = S_1^z \left(S_2^x S_3^y - S_2^y S_3^x \right)$$

$$+ S_2^z \left(S_3^x S_1^y - S_1^x S_3^y \right)$$

$$+ S_3^z \left(S_1^x S_2^y - S_1^y S_2^x \right)$$

For any two spins, $S_n^x S_m^y - S_n^y S_m^x =$

$$= \frac{1}{4i} \left((S_n^+ + S_n^-)(S_m^+ - S_m^-) - (S_n^+ - S_n^-)(S_m^+ + S_m^-) \right)$$

$$= \frac{1}{2i} \left(S_n^- S_m^+ - S_n^+ S_m^- \right)$$

$$\Rightarrow \quad 2i\hat{\chi} = S_1^z \left(S_2^- S_3^+ - S_2^+ S_3^- \right)$$

$$+ S_2^z \left(S_3^- S_1^+ - S_3^+ S_1^- \right) + S_3^z \left(S_1^- S_2^+ - S_1^+ S_2^- \right)$$

~~Thus~~ $\cancel{2i\hat{\chi} \left| \tfrac{1}{2}, \tfrac{1}{2}; \lambda \right\rangle = \frac{2i}{\sqrt{3}} \Big\{ \tfrac{1}{2}(-e^{i\varphi_\lambda}) |\uparrow\uparrow\downarrow\rangle}$

$\cancel{+ \tfrac{1}{2} e^{i2\varphi_\lambda} |\uparrow\downarrow\uparrow\rangle + \tfrac{1}{2}(+|\uparrow\uparrow\downarrow\rangle) + \tfrac{1}{2} e^{i2\varphi_\lambda}(-|\downarrow\uparrow\uparrow\rangle)}$

$\cancel{+ \tfrac{1}{2}(-|\uparrow\uparrow\downarrow\rangle) + \tfrac{1}{2} e^{i\varphi_\lambda} |\downarrow\uparrow\uparrow\rangle \Big\} = \frac{i}{\sqrt{3}} \Big\{ (e^{i\varphi_\lambda} - e^{i2\varphi_\lambda}) |\downarrow\uparrow\uparrow\rangle}$

$\cancel{+ e^{i2\varphi_\lambda} |\uparrow\downarrow\uparrow\rangle - e^{i\varphi_\lambda} |\uparrow\uparrow\downarrow\rangle \Big\}}$

$$2i\hat{\chi}\left|\tfrac{1}{2},\tfrac{1}{2};\lambda\right\rangle = \frac{1}{\sqrt{3}}\Big\{\tfrac{1}{2}|\uparrow\uparrow\downarrow\rangle + \tfrac{1}{2}\left(-|\uparrow\downarrow\uparrow\rangle\right)$$

$$+\tfrac{1}{2}e^{i\varphi_\lambda}\left(-|\uparrow\uparrow\downarrow\rangle + |\downarrow\uparrow\uparrow\rangle\right) + \tfrac{1}{2}e^{i2\varphi_\lambda}\left(|\uparrow\downarrow\uparrow\rangle - |\downarrow\uparrow\uparrow\rangle\right)\Big\}$$

$$= \frac{1}{2\sqrt{3}}\Big\{\left(e^{i\varphi_\lambda} - e^{i2\varphi_\lambda}\right)|\downarrow\uparrow\uparrow\rangle + \left(e^{i2\varphi_\lambda} - 1\right)|\uparrow\downarrow\uparrow\rangle$$

$$+\left(1 - e^{i\varphi_\lambda}\right)|\uparrow\uparrow\downarrow\rangle\Big\} =$$

$$= \frac{1}{2\sqrt{3}}\Big\{\left(e^{i\varphi} - e^{i2\varphi}\right)|\downarrow\uparrow\uparrow\rangle + \left(e^{i\varphi_\lambda} - e^{i2\varphi_\lambda}\right)e^{i\varphi_\lambda}|\uparrow\downarrow\uparrow\rangle$$

$$+\left(e^{i\varphi_\lambda} - e^{i2\varphi_\lambda}\right)e^{i2\varphi_\lambda}|\uparrow\uparrow\downarrow\rangle\Big\} =$$

$$= \frac{e^{i\varphi_\lambda} - e^{i2\varphi_\lambda}}{2}\left|\tfrac{1}{2},\tfrac{1}{2};\lambda\right\rangle$$

$\Rightarrow$ eigenvalues of $\hat{\chi}$ are $\frac{1}{4i}\left(e^{i\varphi_\lambda} - e^{i2\varphi_\lambda}\right)$

$$= \frac{1}{4i}\left(e^{\pm i\frac{2\pi}{3}} - e^{\pm i\frac{4\pi}{3}}\right) = \frac{1}{4i}\left(e^{\pm i\frac{2\pi}{3}} - e^{\mp i\frac{2\pi}{3}}\right) = \frac{\sin\left(\pm\frac{2\pi}{3}\right)}{2}$$

$$\hookrightarrow \frac{4\pi}{3} = \frac{6\pi - 2\pi}{3}$$

States $\left|\tfrac{1}{2},\tfrac{1}{2},\lambda=\pm 1\right\rangle$ have chiralities

$$\chi_\lambda = \frac{1}{2}\left(\pm\frac{\sqrt{3}}{2}\right) = \frac{\sqrt{3}}{4}\lambda \quad (\lambda = \pm 1).$$